

Weekly Learning Debrief

MIS380 — Data Analytics Certificate | Week 5

Topics: Histogram skewness, prediction intervals, regression uncertainty, Excel Forecast Sheet

Discussion: Histogram Skewness

Worked through nine real-world examples, reasoning about left/right/symmetric skew from first principles rather than memorized answers. Core recurring logic: a distribution skews right when there's a floor but no ceiling (income, housing) — most values cluster low, a rare few stretch the tail far to the right. It skews left when there's a ceiling but effectively no floor in the relevant range (an easy exam — most score near the max, a rare few score low).

Example	Skew	Why
Length of hospital stay	Right	Most stays short, rare long stays stretch the tail
Birth weight	Symmetric	Clusters around a healthy average, balanced tails
Family income	Right	Floor near \$0, no ceiling — few very high earners stretch the tail
Daily stock returns	Symmetric	Small gains/losses roughly balanced day to day
Age of death	Left	Most deaths cluster in 80s–90s, tail toward rarer younger deaths
Female shoe size	Symmetric	Clusters around an average, balanced tails
Scores on an easy exam	Left	Most score high, tail toward rarer low scores
Housing prices	Right	Same logic as income — floor near \$0, no ceiling
Adult male height	Symmetric	Classic bell-curve distribution

- Corrected assumption: housing prices, initially guessed as symmetric, are actually right-skewed — same “floor near \$0, no ceiling” logic as income. A small number of ultra-expensive homes stretch the tail, even though most homes cluster in a normal affordable-to-mid range.

Prediction Intervals & Uncertainty Visualization

Modified two charts to visually convey prediction uncertainty, keeping each original chart intact. Both tasks reinforced the same core idea: a model's confidence is not constant — it depends on where a prediction falls relative to the data it was built from.

- **Simple linear regression (restaurant sales vs. population):** added Lower/Upper 95% P.I. as two new line series layered onto a copied chart. The interval was narrowest near the average population value in the dataset (~14) and widened symmetrically toward both extremes — a model is most confident near the center of its training data, least confident at the edges.
- **Time series forecast (stock price):** used Excel's built-in Forecast Sheet tool (Data tab), which generates a forecast plus live-formula confidence bounds via `FORECAST.ETS` and `FORECAST.ETS.CONFINT` — not something to build by

hand from a described formula. Produced the classic “cone of uncertainty” shape: narrow right where the forecast begins, fanning out wider the further into the future it extends.

- **Core takeaway:** a 1-day-ahead forecast is meaningfully more reliable than a 10-day-ahead forecast from the same model — uncertainty compounds with forecast distance, which is why the interval must widen to maintain the same 95% confidence level further into the future.

Notes Worth Remembering

- Excel's Forecast Sheet (Data tab) is the built-in tool for time series forecasting with confidence bounds — not something obvious to discover without being shown, but genuinely useful going forward for any future forecasting task.
- A static, typed-in number in a cell (no formula in the formula bar) can't be explained or defended if asked — worth rebuilding with a live formula/tool rather than leaving unexplained values in a submitted file.
- “Keep the original chart” instructions are satisfied by duplicating the chart (or working on a new sheet) rather than editing the original in place.

Next steps: bring this same format back each week to track growth toward the Mod 8 goal of working independently.